INDIVIDUAL CONTRIBUTION REPORT STUDENT- 1

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Group number	Group 05
Member Role	Student - 1
Module	3D Interactive Media
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GitHub	https://github.com/rashennimsara/TreasureIsland-GroupProject/tree/Member-1-Environment

1. Introduction and project overview

Island Treasure Hunter is a group-developed 3D LEGO-style adventure set on a tropical island. The project combines an explorable environment, character movement and interactive obstacles to support a treasure-hunting experience. Individual members worked on separate systems before bringing their Unity assets into a shared repository. This report records my own contribution, the development choices behind it, collaboration, testing and areas that require further verification. It distinguishes prepared components from fully tested gameplay so that the evidence accurately reflects my work during CW1.

2. My role and responsibilities

My responsibility in Island Treasure Hunter is to develop the LEGO-style island environment and implement the LEGO Brick Rigidbody Physics Controller. So far, I have shaped the island and coastline, built the LEGO-style beach, ocean and grass areas, and added the ocean, mountain, waterfall, river, rocks, trees and bushes. I used plain colour materials to give the environment a consistent LEGO look, added a Directional Light to cast shadows, and used the same animated URP water material for the sea and river. I also laid out the exploration route from the beach through the forest towards the river and mountain, leaving enough space for the player to move and jump.

My next task is to build LEGO stepping stones and add obstacles across the river. For the Rigidbody controller, I plan to add a Rigidbody and a Collider to one LEGO brick and write a script that applies a force when the player lands on it. I will test how the brick moves, how it collides with the player, and whether the mission fails as expected if the player falls into the river **(Unity Technologies, n.d.).**

LEGO Minifigure Movement and Jumping (Per Student)

I set up a playable LEGO minifigure in Unity. The player uses the W, A, S and D keys to move around the scene and the space bar to jump. The player character currently appears as a capsule, and its movement and jump controls are functional.

Assigned Area	My contribution / current status
Island and ground	Create Circular island, collider, sand and green LEGO-style tiles
Water and terrain	Create Ocean shader, mountain, waterfall and river with waves
Environment props	Trees, palms, bushes and rocks arranged for visibility
Route planning	Create a route (beach to forest)

3. Development and implementation

3.1 Island shape and ground composition

I created a circular to give the level a more natural circular footprint. Its scale was set to 400, 0.5, 400, and I configured a Mesh Collider to correspond with the island surface. This provided a large foundation for the rest of the environment and a surface that could be checked during character integration. I then placed LEGO-style ground tiles across the island. Sand materials were applied around the coast, while green ground materials covered the interior. This contrast was intentional: the sand clearly communicates the beach and starting area, while the green centre suggests a forest and exploration space. I arranged the tiles to maintain a consistent LEGO-inspired visual style and to avoid an entirely flat, featureless environment. The island's shape and ground treatment were important because the group's interactive systems needed a recognisable setting rather than isolated test objects.

3.2 Ocean, waterfall and river

I expanded the ocean around the island to make the level feel surrounded by water rather than bordered by empty space. I used an animated URP water shader to create moving wave patterns and highlights **(Ilett, 2020)**. This was a visual feature, so it needed to remain compatible with the project's render pipeline and scene lighting. I also added a rocky mountain with a green upper surface, an animated waterfall, a splash pool and a winding river flowing towards the sea. For the river, I lowered the channel and checked the surrounding ground colliders so the water could occupy a visible path. I hid ground tiles that overlapped the channel, preventing the river from being visually covered by the LEGO terrain. These changes gave the environment a stronger sense of direction: the mountain acts as a landmark, while the river introduces a natural division in the player's route. The river crossing itself was left for the teammate responsible for stepping-stone obstacles.

3.3 Trees, rocks and visual organization

I placed LEGO-style broadleaf trees, palm trees, bushes and rocks in clusters around the island. I avoided filling every available space, because the game requires room for the player to walk, turn and interact with objects. I kept important features such as the waterfall, river and coastline visible rather than hiding them behind dense vegetation. The vegetation also helps communicate different areas: palms support the beach setting, while broader trees and bushes suggest the island interior. Rocks provide visual variety and help frame the mountain and river. During placement, I considered the camera view and the possibility that teammates would later add interactive objects. The environment was therefore arranged as a playable space in progress, not only as a decorative scene.

3.4 Exploration route and scene planning

Before arranging the walking route, I referred to visual concept images to consider how a player might travel through the environment. The planned path begins near the beach, winds through vegetation and reaches one riverbank. It then resumes on the opposite bank and continues through the forest towards a finishing area behind the mountain. This design gives the player a sequence of recognizable locations to follow. The river acts as a planned obstacle and a transition between parts of the level. I left space at the river crossing for the stepping stones that I plan to build myself in CW2. This approach also helped me understand the need to coordinate scene layout with other members before every interaction was complete.

4. Collaboration, GitHub and integration

I used GitHub Desktop to track and commit my environment work on the Member-1-Environment branch. The work included Unity scene changes, editor scripts, shaders, materials, generated meshes and related .meta files. Keeping these files together matters because Unity uses metadata to maintain asset references. I worked on the environment while other members developed movement, mouse interaction and puzzle systems. The shared repository provided a way to transfer the island work for integration without sending separate, disconnected assets. The scene was prepared for combining with the other members' features; however, final gameplay behaviour should be confirmed in the combined project rather than assumed from the environment scene alone.

5. Testing, challenges and solutions

I reviewed the overall island shape, material separation and placement of the main landscape features in the Unity scene. I adjusted the river channel and hid overlapping ground tiles so that the water remained visible. I also considered whether trees, rocks and bushes obstructed the planned route, leaving open areas for movement. The most important integration checks are ground collision, player clearance, water behaviour and camera visibility once the final character and gameplay systems are attached. A visual inspection of the environment is not the same as a complete play test.

6. Individual reflection and future improvements

This contribution improved my understanding of scene composition, material assignment, colliders and how environment design affects interaction. A major lesson was that decorative details must support navigation instead of making movement confusing. The river and mountain also showed me how landmarks can guide players through a level. Using GitHub reinforced the importance of keeping scene assets and their metadata organised for group work. If I continued this work for CW2, I would prioritise testing the route with the final character, refining the river crossing with the assigned teammate and checking the performance of animated water and repeated environment props.

7. References

Ilett, D. (2020) Stylised water in Shader Graph and URP. Available at: https://danielilett.com/2020-04-05-tut5-3-urp-stylised-water/ (Accessed: 10 November 2026).

OpenAI *ChatGPT* [Generative AI tool]. Available at: https://chatgpt.com/ (Accessed: 10 November 2026).

Unity Technologies (n.d.) *Rigidbody component reference*. Unity 6.0 User Manual. Available at: https://docs.unity3d.com/6000.0/Documentation/Manual/class-Rigidbody.html (Accessed: 10 November 2026)

DitzelGames (2019) *Making a LEGO® Game in Unity - LegoGame 1* [YouTube video]. 3 August. Available at: https://www.youtube.com/watch?v=43diUOkhA-k (Accessed: 10 November 2026).

iHeartGameDev (2021) *Intro to Game Development with Unity | Lego Microgame Setup & Walkthrough [#1]* [YouTube video]. 21 February. Available at: https://www.youtube.com/watch?v=62d5ZHo-pNc (Accessed: 10 November 2026).

8. AI assistance and responsible use

No	Question	What AI gave
1	How can I light the island and show shadows from its objects?	Use a Directional Light as the main daylight source. Adjust its angle and check the shadows from rocks, trees and other LEGO props in Game View.
2	How can I stop LEGO ground tiles from covering parts of the river?	Move or remove the tiles that overlap the river channel. Check the result from the player's camera to make sure the water and riverbanks are visible.
3	How can I make the walking route clear in a LEGO-style environment?	Use the same animated URP water material for both surfaces. This gives the sea and river the same wave effect
4	How can I give the sea and river a consistent appearance?	Use the same animated URP water material for both surfaces. This gives the sea and river the same wave effect.

9. References and asset credit

Appendix A — Evidence of individual contribution



Figure 1: Circular island with sand coastline , green LEGO ground and LEGO Minifigure Movement and Jumping

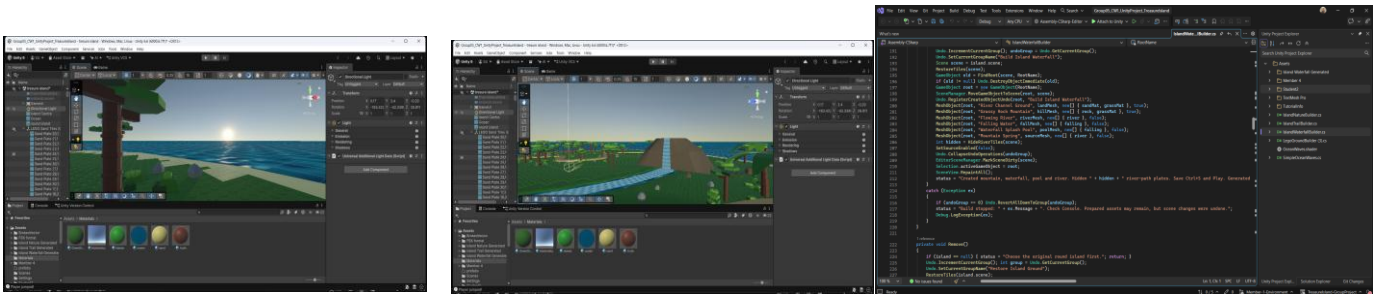


Figure 2: Ocean, mountain, waterfall and winding river

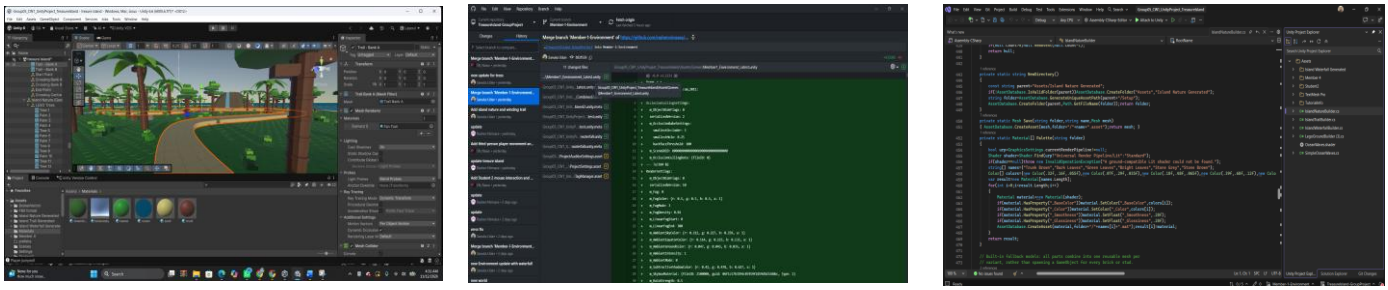


Figure 3: Trees, route layout and Member-1-Environment GitHub commits

Appendix B — Additional supporting records

AI chat transcripts

